Molecular Catalysis 535 (2023) 112852

Contents lists available at ScienceDirect

Molecular Catalysis

journal homepage: www.journals.elsevier.com/molecular-catalysis

First-principles study of oxygen evolution on Co_3O_4 with short-range ordered Ir doping

Zhuang Zhang [a], Guoying Tan [a], Anuj Kumar [b], Hai Liu [a], Xue Yang [a], Wenqin Gao [a], Lu Bai [c], Huaiqiu Chang [c], Yun Kuang [a], Yaping Li [a,*], Xiaoming Sun [a]

[a] State Key Laboratory of Chemical Resource Engineering, College of Chemistry, Beijing University of Chemical Technology, Beijing 100029, China
[b] Department of Chemistry, Institute of Humanities and Applied Science, GLA University, Mathura 281406, India
[c] CAS Key Laboratory of Standardization and Measurement for Nanotechnology, National Center for Nanoscience and Technology, Beijing 100190, China

ARTICLE INFO

Keywords:
First-principles
Oxygen evolution reaction
O/OH coverage
Short-range ordered
Solvation effect

ABSTRACT

The large overpotential of oxygen evolution reaction (OER) in water-splitting process prevented its application in energy field, researchers attempted to find efficient catalysts. Here, we used short-range ordered Ir-doped Co_3O_4 (100) model to study the OER performance under various coordination conditions through first-principles calculations. This doping pattern enabled the model to display three types of active sites, included single Ir/Co site and double Ir sites, which were different in terms of propensity to adsorb reaction intermediates. The OER performance of three types of active sites with the same O and OH coverage was evaluated, we found that O covered surface increased OER catalytic activity for single Ir/Co site. While for double Ir sites, it exhibited the lowest overpotential among the above three types, and O/OH coverage had a little influence on its activity. Furthermore, Bader charge and partial density of states analyses showed that, for single Ir/Co site, the overpotential on covered surface was reduced due to the valence electrons decrease at the metal site. While for double Ir sites, the overpotential reduction was due to the unique adsorption patterns. Additionally, the solvation effect was considered. It was found that the solvation had a key effect on the adsorption energy depending on the binding site of the intermediate. But the overpotential trend in solvent was similar with that in vacuum on the same active site.

Introduction

With the depletion of fossil fuels expected to worsen, it was crucial to find alternative, renewable, and sustainable energy resources to fulfill the growing energy demand for economic growth [1,2]. In this regard, hydrogen production, a viable energy source due to its high energy density and pollution-free usage, via electrochemical water splitting can be considered an efficient strategy [3–5]. However, because the extremely sluggish kinetics of oxygen evolution reaction (OER) at the anode, overall water splitting was hindered at a practical level [6]. Currently, IrO_2 was the only known industrial OER electrocatalyst [7–10]. However, it was difficult to implement large-scale production and use of such catalysts because precious metals were inherently scarce and expensive (iridium US$60,670 kg^{-1}) [11]. It was therefore crucial to develop efficient and inexpensive electrocatalysts to reduce the dependence on precious metals.

First-row transition metal oxides were much more abundant than precious metals, and they had exhibited significant potential as OER catalysts [12–16]. Particularly Co and its oxides or hydroxides can effectively promote electrochemical water splitting, and it also had certain stability under acidic, alkaline, and neutral conditions [17,18]. The low price of Co_3O_4 made it possible for a wide range of applications (Co_3O_4 US$2706 kg^{-1}) [19]. Recently, the Galán-Mascarós [20] group used hydrophobic carbonyl substrates to process Co_3O_4 catalysts and found that it could avoid proper solvation of the electrode surface and prevent metal oxide dissolution. The Co_3O_4@C (carbon) could work stably for more than 40 h at 1 M H_2SO_4 under 10 mA cm^{-2} current density. Bhatti et al. [21] grew Co_3O_4 nanostructures using organic polyelectrolytes and polyacrylamide, and the resulting Co_3O_4@CMC (Carboxymethyl cellulose) composites had an OER overpotential of 290 mV at 10 $mAcm^{-2}$ in 1.0 M KOH. Cai et al. [22] studied Ir single atoms on amorphous CoO_x nanosheets, which together could initiate the OER at a low onset potential below 30 mV. These materials achieved superior OER activity, which exceeded that of commercial IrO_2 by 160 times.

* Corresponding author.
E-mail address: liyp@mail.buct.edu.cn (Y. Li).

https://doi.org/10.1016/j.mcat.2022.112852
Received 17 August 2022; Received in revised form 13 November 2022; Accepted 29 November 2022
Available online 6 December 2022
2468-8231/© 2022 Elsevier B.V. All rights reserved.

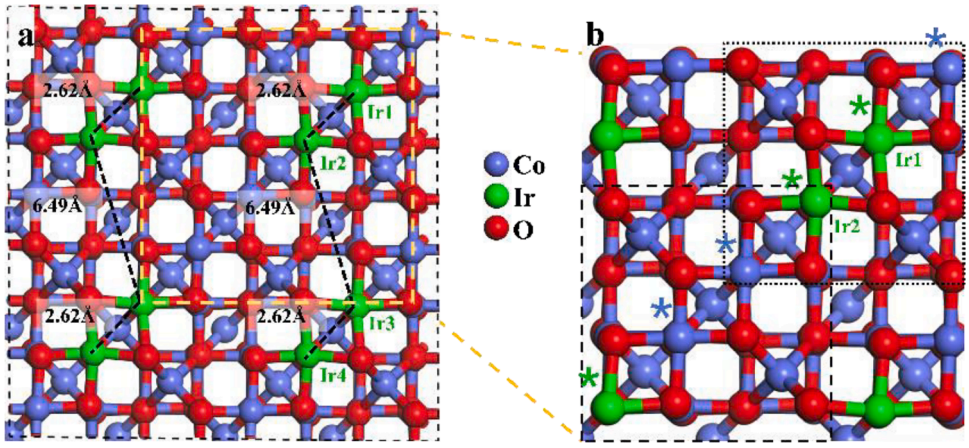


Fig. 1. Top view of short-range ordered Ir doped Co_3O_4 (100). (a) top view (b) schematic diagram with coverage/adsorption sites.

Table 1
Energy, ZPE, and entropy (eV) for gaseous and adsorbed molecules.

	E	ZPE	TS
H_2(g)	−6.76	0.347	0.40
O_2(g)	−8.81	0.134	0.63
H_2O(l)	−14.22	0.570	0.67

Doping of noble metals was an effective means to enhance the OER activity of Co_3O_4 substrates. Qiao et al. [23] observed that short-range ordered Ir adopted a special electronic structure that was different from that of Ir single atoms and pure Ir nanoparticles. Ir was more prone to doping at Co(III) sites in octahedral units than at Co(II) sites in tetrahedral units. Specifically, the $Ir_{0.06}Co_{2.94}O_4$ material achieved excellent catalytic OER activity and corrosion resistance. Although previous studies had reported strategies to enhance Co_3O_4 by noble metal doping, there were only a few reports that mention the influencing factors of the surface four-electron process of OER in short-range ordered Ir doping. The Nørskov group calculated the OER theoretical overpotential based on ΔG(*O), ΔG(*OH), and ΔG(*OOH) [24]. Here, *OH, *O and *OOH were all intermediates of the OER, and all three species can be adsorbed on the catalyst surface. Rossmeisl et al. [25] confirmed that *O and *OH exhibited more stable adsorption. Our group's previous research indicated that the OER activity of Co sites was enhanced when the surface was covered with a 1/2 monolayer (ML) O or 1/2 ML OH [26]. The Bader charge indicated that Co sites had more valence electrons when the surfaces were covered with O or OH moieties. Different range coverage of various intermediates also has a significant impact on the OER performance of NiFe-LDHs [27,28]. In general, the amount of O and OH covered on the surface of a catalyst affected the OER mechanism and made it harder to find the active sites. Furthermore, solvation effects were particularly important for electrochemical reactions that occur at the electrode-electrolyte interface (water solvent: $\varepsilon = 78.4$) [29].The implicit solvation was the method to simulate the solvation effect by continuous medium response [30]. Zhang et al. [31] used vaspsol to study the ORR performance of the Pt (111) surface, which improved the vacuum calculations to a certain extent.

Previous experimental studies have shown that spinel Co_3O_4 synthesized by various methods was mainly exposed to (100) and (111) surfaces and rarely terminated at (110) surfaces [32,33]. Indeed, the theoretical calculation also verified that the (100) surface of spinel Co_3O_4 has better stability than the (111) and (110) surfaces [34]. In addition, the surface of spinel Co_3O_4 (111) was still covered by water molecules even at high temperature (>450 °C) due to the highly unsaturated coordination Co atoms, and only surface (100) was the most abundant surface of catalytic activity [35]. Therefore, we selected the (100) surface of spinel Co_3O_4 to study, which was consistent with the previous experimental and theoretical reported. The mass ratio of Ir in the model was 17.5%, so the cost of the model was about US$12,850 kg^{-1}. The lower price made it have better application prospects. Short-range order referred to ordering at distances within the radius 10–12 Å region [36,37]. Zhu's group [38] proposed that the characteristic length of short-range order was the distance of a several atoms. Qiao et al. [23] thought that Ir atom was located at the octahedral position point of Co_3O_4, observed that the spatial correlation between Ir substitution sites and Co_3O_4 host lattice, and these Ir single atoms showed a short-range order. Xu et al. [39] confirmed Co(III) was dominant active sites of spinel Co_3O_4 for OER. While in our calculation model, we choose the (100) surface of Co_3O_4 and exposed with Co(III) atoms. The distance of the adjacent Ir atom was 2.62 Å or 6.49 Å by replaced Co(III) atoms. It was found that the Ir sites displayed short-range order rather than exist as isolated sites within Co_3O_4, as shown in the Fig. 1(a). The square of dotted line region and dashed line region in Fig. 1(b) were chosen to investigate the effect of same O and OH coverage on the performance of different active sites.

The present work main focused on (i) the OER performance of Ir doped Co_3O_4 (100) with short-range order (the distance of the adjacent Ir atom was 2.62 Å or 6.49 Å), (ii) the OER performance with the same O/OH covered surface on the different metal sites. Then, we calculated electronic density of states and Bader charge of the above sites with the same O/OH coverage. Finally, we considered the solvation effect to evaluate the OER performance on different sites.

Computational models and theoretical methods

For this study, we performed density functional theory (DFT) calculations using a projector augmented wave method in the Vienna Ab-initio Simulation Package (VASP) program [40,41]. The exchange and correlation energy density functions were computed using the Perdew-Burke-Ernzerhof (PBE) generalized gradient approximation (GGA) [42,43]. The DFT+*U* method was used to optimize the geometric and electronic structures, where U value was 3.5 for Co [44,45]. The optimized lattice parameters a, b, c, α, β, and γ were 8.08, 8.08, 17.20 Å, 90°, 90°, and 90°, respectively, and the Co_3O_4 (100) model consisted of eight Co layers; all of these parameters were similar to those found in a previous study [46]. When performing thermodynamic calculations, the four bottom layers were fixed, and the four upper layers were released. The Monkhorst-Pack-type k-point sampling parameters were $5 \times 5 \times 5$ and $3 \times 3 \times 1$ to optimize the bulk structure and slab model, respectively. The cutoff energy was 400 eV, consistent with our group's previous work [26]. The convergence criteria for force and energy were 0.05 GPa and 10^{-4} eV/Å, respectively. The bulk spin and dipole correlation were not restricted, and we determined the atomic charge density

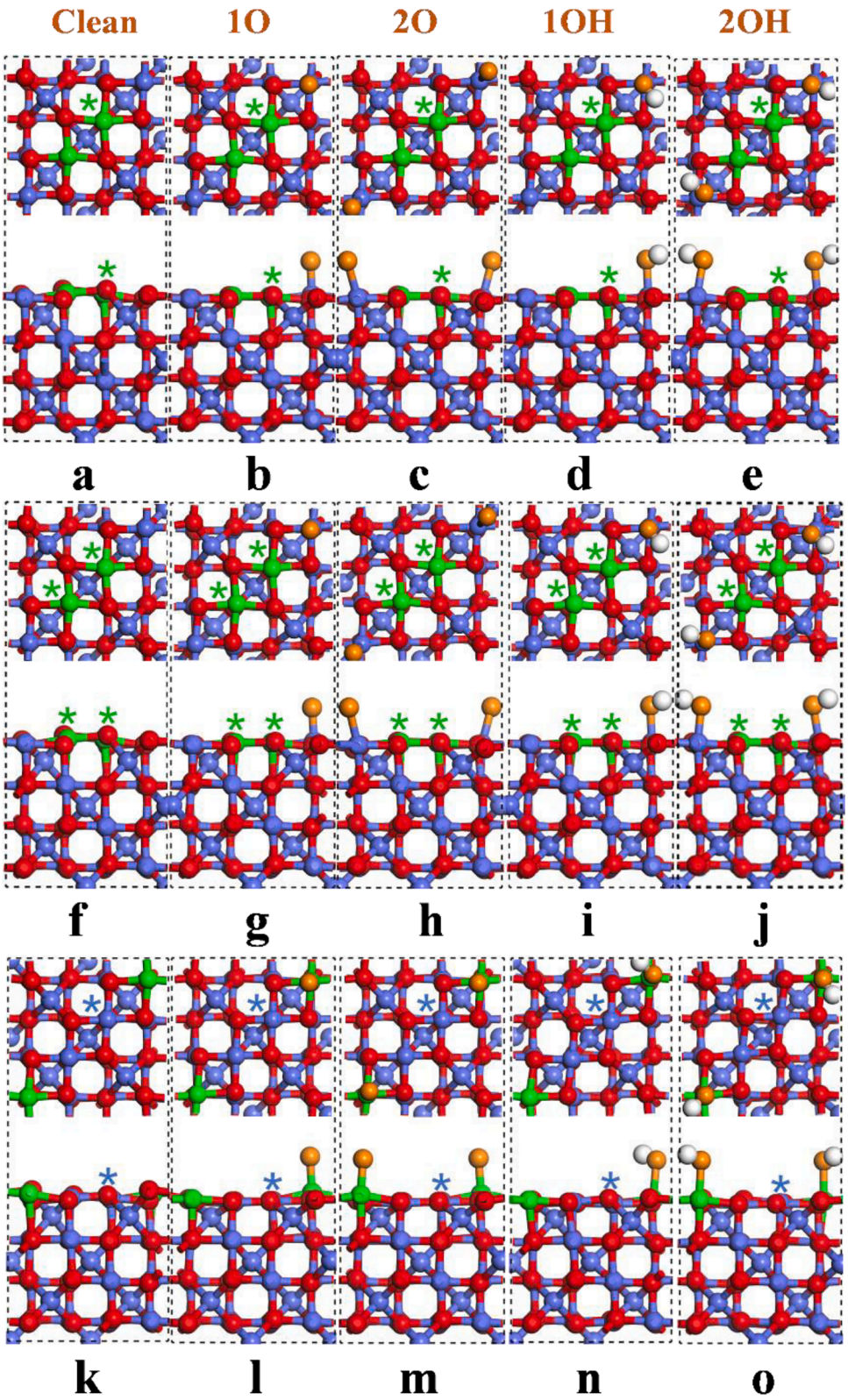


Fig. 2. Top and side views of O and OH coverage on different sites on the surface of Co_3O_4 (100) with short-range ordered Ir doping: (a-e) single Ir site with various O and OH adsorption cases; (f-j) double Ir sites with the same O and OH adsorption cases; (k-o) single Co site with the same O and OH adsorption cases. The active site was marked with "*". Co, Ir, covered O, other O, and H appeared in blue, green, orange, red, and white, respectively.

corresponding to the adsorbed species based on Bader charge analysis [47]. The distances between two Ir atoms were arranged in a periodic pattern of 6.49 Å and 2.62 Å, and Ir atoms were found in octahedral sites [23]. In addition, vaspsol continuum solvation model was used to evaluate the solvation effect of the OER performance, which was one of several available procedures for simulating solvation energies of periodic models [31]. Previous studies have shown that short-range ordered Ir doping at this distance exhibits better OER activity. However, to our

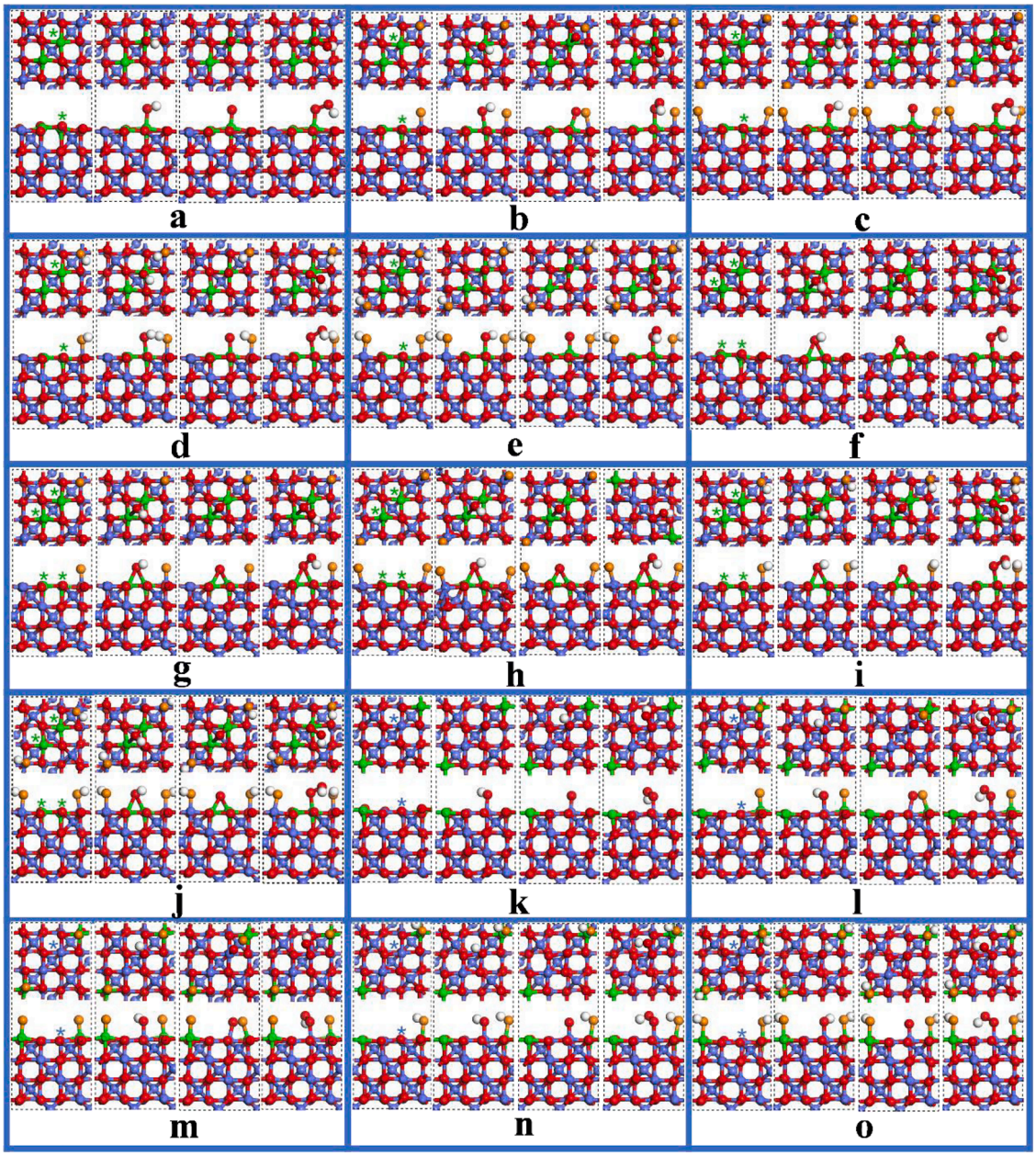


Fig. 3. (a-e) Top and side views of optimized structures of all adsorbed intermediates on single Ir site with various O and OH covered; (f-j) double Ir sites with the same O and OH covered; (k-o) single Co site with the same O and OH covered. Oxygen atoms participated in OER were red, and covered oxygen atoms were orange.

knowledge, the effects of different coordination conditions on the surface catalytic activity have not been studied.

The overpotential of the OER was calculated according to the simplified scheme developed by Nørskov et al. [24]. The OER involved four proton-coupled electron transfer steps, where an electron was transferred to the electrode and a proton was transferred to water, as expressed in Eq. (1–5):

$$H_2O + * \rightarrow *OH + H^+ + e^- \quad (1)$$

$$*OH \rightarrow *O + H^+ + e^- \quad (2)$$

$$H_2O(l) + *O \rightarrow *OOH + H^+ + e^- \quad (3)$$

$$*OOH \rightarrow * + O_2 + H^+ + e^- \quad (4)$$

In these equations, *denoted an active site, and thus, *OH, *O, and *OOH indicated adsorbed intermediates. The Gibbs free energies for the relevant species were calculated relative to the standard hydrogen electrode (SHE) for each step, as shown in Eq. (5-11),

$$\Delta E(OH) = E(*OH) - E(*) - [E(H_2O) - 0.5(H_2)] \quad (5)$$

$$\Delta E(O) = E(*O) - E(*) - [E(H_2O) - E(H_2)] \quad (6)$$

$$\Delta E(OOH) = E(*OOH) - E(*) - [2E(H_2O) - 1.5E(H_2)] \quad (7)$$

$$\Delta G(1) = \Delta E(OH) + \Delta ZPE - T\Delta S - eU + k_B T \ln 10 \cdot pH \quad (8)$$

$$\Delta G(2) = \Delta E(O) + \Delta ZPE - T\Delta S - eU + k_B T \ln 10 \cdot pH \quad (9)$$

$$\Delta G(3) = \Delta E(OOH) + \Delta ZPE - T\Delta S - eU + k_B T \ln 10 \cdot pH \quad (10)$$

$$\Delta G(4) = 4.92 - \Delta G(1) - \Delta G(2) - \Delta G(3) \quad (11)$$

where E(*X) and ΔE(*X) represented the total energy and adsorption

Table 2
Calculated free energies (eV) and overpotentials (V) for the evaluated structures. η(sol) (V) was the overpotential of each structure with solvation.

Structures	ΔG(1)	ΔG(2)	ΔG(3)	ΔG(4)	η	η(sol)
a	0.38	1.07	2.04	1.43	0.81	0.87
b	0.21	1.52	1.82	1.37	0.59	0.47
c	0.28	1.05	2.05	1.54	0.82	0.88
d	0.19	1.76	2.14	0.82	0.91	0.81
e	0.11	1.66	2.15	1.00	0.92	0.97
f	0.16	1.54	1.63	1.59	0.40	0.72
g	0.15	1.66	1.69	1.41	0.46	0.79
h	0.23	1.61	1.72	1.36	0.49	0.75
i	0.06	1.65	1.69	1.51	0.46	0.72
j	0.26	1.66	1.68	1.31	0.45	0.71
k	0.59	2.29	0.86	1.16	1.06	0.82
l	1.15	1.26	1.89	0.62	0.66	0.44
m	1.16	1.03	2.06	0.67	0.83	0.57
n	1.07	2.30	0.80	0.74	1.07	1.03
o	1.06	2.51	0.80	0.55	1.28	1.09

energy of intermediate species X (X = OH, O, and OOH), respectively, as calculated using DFT. We used the energy of 0.5 H_2 instead of the energy of $[H^+ + e^-]$ at a pH = 0, P = 1 atm, and T = 298 K [48]. The changes in energy (ΔE), zero point energy (ΔZPE), and entropy (TΔS) for H_2, O_2, and H_2O under the same conditions are presented in Table 1 [49]. Then, we performed single-point calculations including implicit solvent effects (water solvent: ε = 78.4), E(sol) was the solvation energy calculated using the continuum model, the total energy includes the gas phase energy and solvation energy [31]:

$$\mathrm{E(total)} = \mathrm{E}(gas) + \mathrm{E}(sol) \tag{12}$$

ΔE(sol) was the sum of E(sol) and ΔE and represented the binding energy in solvation, ΔE was the binding energy in vacuum [50].

$$\Delta E(sol) = E(sol) + \Delta E \tag{13}$$

The terms "e" and "U" in the equations above represented an electron and the electrode potential vs. SHE, respectively; therefore, $-eU$ was the free energy change when transferring one electron. The effect of different pH conditions on the overpotential can be corrected by introducing the factor "$k_B T ln10 \cdot pH$", where k_B was Boltzmann's constant [24]. We used the $4e^-$ mechanism for the DFT calculations, which was consistent with the scheme developed by Nørskov's group [25]. Here, ΔG(1–4) were the free energy changes corresponding to each electron transfer step, and ΔG(4) was calculated as 4.92−ΔG(1)−ΔG(2)−ΔG(3), because 4.92 eV was the theoretical free energy change for the reaction, $O_2 + 4(H^+ + e^-) \rightarrow 2H_2O$. The largest value among ΔG(1–4) indicated the potential-determining step, and the η = max[ΔG(1), ΔG(2), ΔG(3), ΔG(4)]/e − 1.23 V.

Results and discussion

In this study, DFT+U calculations were performed to study the OER performance of different active sites on the optimized surface affected by the same O and OH surface coverage. This special Ir atoms distance enabled the intermediates to be adsorbed by two adjacent Ir sites at the same time, so there will be unique double Ir sites on the surface of this model. As shown in Fig.S1, the Ir-Co double sites would relax to single Ir site, which was due to the different adsorption capacity of oxygen to the different sites. The oxygen adsorbed at the double Co sites inevitably chemical interacts with the lattice oxygen due to the short Co-O bond length, which was not preferred by the adsorption evolution mechanism (AEM) [51]. The optimized slab structures of fifteen modelled structures were shown in Fig. 2. Each row in Fig. 2 showed the same type of active site, and each column showed the same O and OH coverage state. The three active sites include: single Ir site, double Ir sites and single Co site. The three active sites with respect to O and OH on surface coverage includes five cases: clean surface, one oxygen covered, two oxygen covered, one OH covered, and two OH covered. Other potential O and OH covered cases were shown in Fig.S2.

The four electron-transfer processes at single Ir site, single Co site, and double Ir sites with adsorbed O and OH moieties were illustrated in Fig. 3, and the cases were denoted as structures (a–o). In the first step, a water molecule on a clean surface donated a proton and an electron to made an OH group that remained adhered to the surface. Then, the adsorbed OH deprotonated, left behind an O, which underwent a chemical interaction with another water molecule to produce OOH and a proton. Finally, the surface was restored to its initial condition when the adsorbed OOH released one molecule of dioxygen and one proton.

The involved $4e^-$ process at single Ir site was depicted in Fig. 3(a-e), and the illustrated mechanisms were consistent with those on the pristine catalyst surface.

The OER process was further displayed in Fig. 3(f-j), which revealed that all intermediates will interact with double Ir sites; this reactivity was distinct from that at the traditional single site. The optimized structures of single Co site adsorbed OER intermediates were shown in Fig. 3(k-o). Adsorption energies of OH, O, and OOH affected the free energy change associated with each stage of the OER process. Furthermore, Table 2 summarized the findings of free energy calculations and overpotential with and without solvation effect for each structure. It can be found that the single Co site on the clean surface required the largest free energy to accomplish the change from OH to O.

The free energy diagrams of all structures were presented in Fig. 4. The double Ir sites had the lowest overpotentials among the three types of evaluated sites, and the corresponding free energy plots with different O and OH coverages were shown in Fig. 3(f-j). Overall, the double Ir sites on a clean surface exhibited the best OER activity, and these results were superior to those obtained in our previous calculations for Co_3O_4 without Ir doping [26].

The free energy diagrams for single Ir site with various O and OH coverage and on a clean surface were shown in Fig. 4(a-e), while the free energy diagrams of the single Co site under the same coverage conditions were shown in Fig. 4(k-o). Zeng et al. [52] investigated a single-atom Ir loaded CoOOH substrate to enhance catalytic activity. The calculated results showed that the overpotential was 0.85 V and 0.40 V when the Ir was loaded on the triple hollow position of surface oxygen and the oxygen vacancy, respectively. While in this work, the overpotential data listed in Table 2 confirmed that the single Ir site had lower overpotentials than the single Co site under the same coverage conditions: specifically, Ir-clean (η=0.81 V) < Co-clean (η=1.06 V); Ir-1O (η=0.59 V) < Co-1O (η=0.66 V); Ir-2O (η=0.82 V) < Co-2O (η=0.83 V); Ir-1OH (η=0.91 V) < Co-1OH (η=1.07 V); Ir-2OH (η=0.92 V) < Co-2OH (η=1.28 V). These results highlighted the key role of the noble metal in promoting the transition metal oxide substrate's OER activity [53].

The overpotentials of the single Ir site with various O and OH covered followed the trend: Ir-1O (η=0.59 V) < Ir-clean (η=0.81 V) < Ir-2O (η=0.82 V) < Ir-1OH (η=0.91 V) < Ir-2OH (η=0.92 V). The overpotentials of the single Co site followed a similar trend: Co-1O (η=0.66 V) < Co-2O (η=0.83 V) < Co-clean (η=1.06 V) < Co-1OH (η=1.07 V) < Co-2OH (η=1.28 V). The He's group [54] reported that surface partial oxidized can promote the OER activity of $Ni_{(1-x)}Fe_xP_2$ catalyst. Tηe calculated results showed that the OER performance of the site was improved when the surface has 1/4 oxygen covered [54]. Analyzing these overpotential trends revealed that Ir doping and oxygen adsorption both significantly impacted the OER performance of Co_3O_4. The effect of O and OH covered the surface on single Ir site and single Co site was shown in Fig.S4 (a), the single Ir site and the single Co site simultaneously have lower overpotentials when the oxygen coverage was 1O. However, Fig.S4 (b) showed that the double Ir sites activity was almost unaffected by surface O and OH coverage.

Adsorption energies of the three active sites under the same surface coverage conditions were shown in Table 3. For most structures, the *O

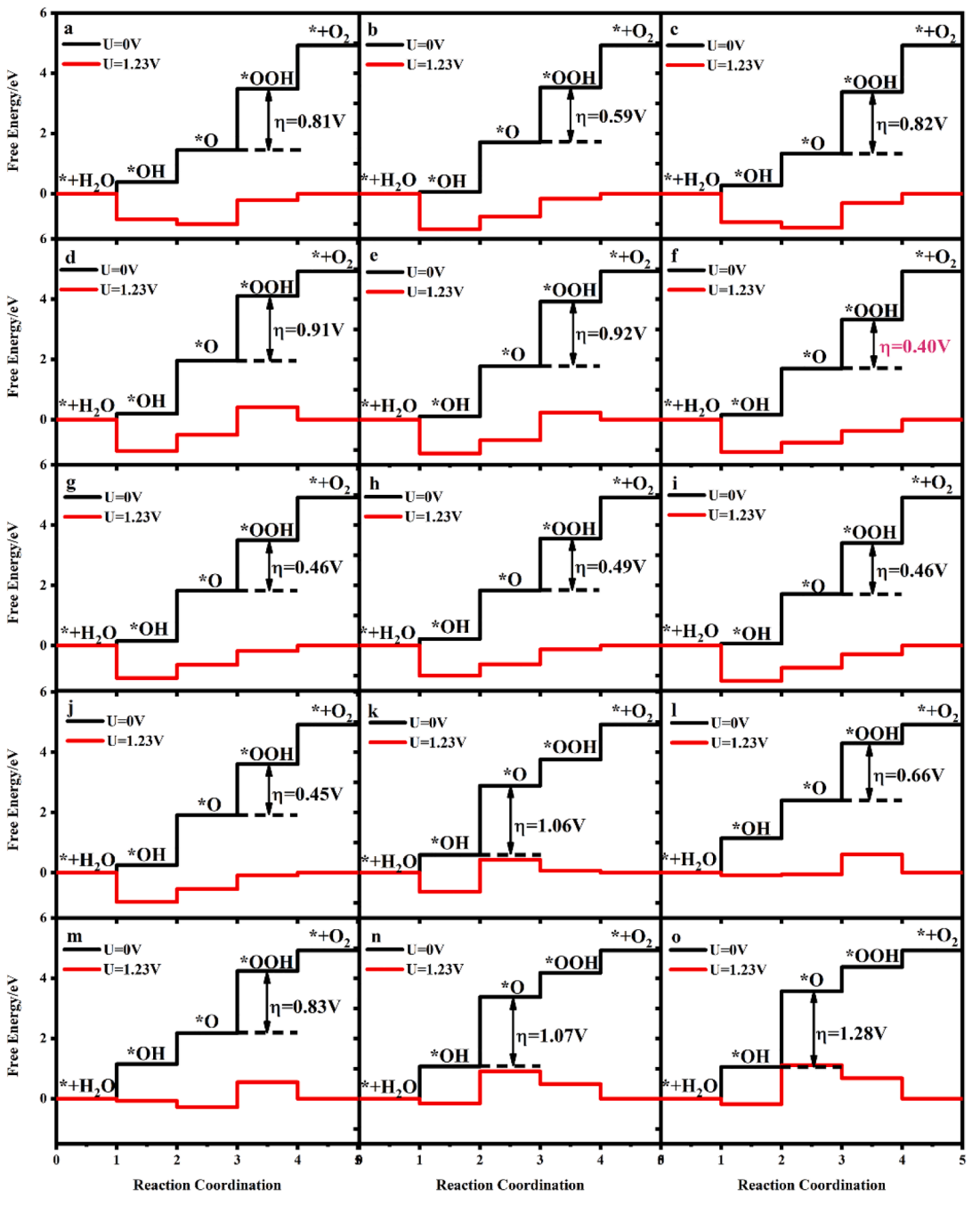


Fig. 4. Free energy diagrams of the four single-electron steps on (a) single Ir site on a clean surface, (b,c) single Ir site of one and two O-covered surface, (d,e) single Ir site of one and two OH-covered surface, (f) double Ir sites on a clean surface, (g,h) double Ir sites on one and two O-covered surface, (i,j) double Ir sites of one and two OH-covered surface, (k) single Co site on a clean surface, (l,m) single Co site of one and two O-covered surface, (n,o) single Co site of one and two OH-covered surface.

to *OOH conversion was the rate-determining step, hence the adsorption energies of O and OOH moieties on the catalyst surface were the primary determinants of the overpotential. Since the OER intermediates adsorption energy existed scaling relations, the minimum theoretical overpotential of adsorbate evolution mechanism (AEM) was 0.37 eV [55]. In addition, the volcano plot included the theoretical maximum value of the adsorbate evolution mechanism (AEM) and the overpotential of various structures, as shown in Fig. 5(a) and Fig. 5(b). Fig. 5 (a) displayed the correlation between ΔE(*O) and −η, suggested that the oxygen binding energies were highest at the site with a single Co atom and lowest at the site with a single Ir atom. Both of two situations were adverse for more reactivity. The optimal adsorption energy for oxygen was reached at the double Ir sites, which was beneficial for the subsequent reaction.

Fig. 5(b) depicted the plot of overpotential and ΔE(*OOH) for multiple sites with varying surface coverage. Here again, the double Ir sites had the optimal OOH adsorption energies compared to the other sites we considered; the single Ir site had significantly lower OOH adsorption energies, while the single Co site had very large OOH adsorption energies, which was not conducive for lowering overpotential. A small overpotential can only be reached by making sure that the sum of all the steps' adsorption energies was optimized as possible. The calculated overpotential values for all structures were below 0.37 V, and the theoretical maximum close to 0.37 V, which was in line with previous

Table 3
Calculated adsorption energies (eV) of all structures.

Structures	ΔE(*OH)	ΔE(*O)	ΔE(*OOH)
a	−0.06	1.34	2.93
b	−0.23	1.62	3.24
c	−0.16	1.23	2.82
d	−0.24	1.86	3.54
e	−0.33	1.67	3.36
f	−0.29	1.60	2.77
g	−0.30	1.72	2.94
h	−0.22	1.74	3.00
i	−0.38	1.62	2.85
j	−0.19	1.82	3.05
k	0.17	2.80	3.20
l	0.71	2.30	3.74
m	0.73	2.07	3.69
n	0.65	3.26	3.62
o	0.63	3.45	3.81

theoretical studies [55]. Fig. 5(c) displayed a linear relationship between the adsorption energies of O and OH across all structures: [ΔE(*OH) = 0.66ΔE(*O) – 1.22 eV], which was consistent with previous results [56]. Meanwhile, the OH and OOH adsorption energies for all structures had a closer linear relationship, [ΔE(*OOH) = ΔE(*OH) + 3.14 eV], as shown in Fig. 5(d), which was also consistent with previous reported [57]. The partial density of states (PDOS) of the three tested active sites under same surface coverage conditions were investigated and compared, as shown in Fig. 6. Fig. 6(a) showed the PDOS of single Co site with varying amounts of O and OH covered revealed that the occupied states below the Fermi level were bonded by Co 3d orbitals. Moreover, the density of states for single Co site were similar regardless of the extent of O and OH coverage, O and OH adsorption had little effect on the electronic states of single Co site. The D-band center of Co on the clean surface was about −1.58 eV and −1.31 eV after one O covered, the d-band center of the single Co site shifted toward the Fermi level. This finding suggested that the intermediate might be adsorbed more securely on the single Co site because more antibonding orbitals contained no electrons.

The DFT simulations showed that adsorption of oxygen on the catalyst surface could bring the Fermi level closer to the Co 3d (εd) band center. The Co 3d band centers for each case followed the order, −1.50 eV (Co/2OH) < −1.40 eV (Co/1OH) < −1.31 eV (Co/1O) < −1.27 eV (Co/2O). The D-band centers of structures l and m were closer to the Fermi level, which suggested that the binding of oxygen intermediates to Co active sites was significantly enhanced. Previous studies have shown that Co atoms can easily adsorb hydroxyl groups [58], thereby making it more difficult to extract a proton from the OH moiety. As a result, the ΔE(*OH) value was small for the single Co site on the clean surface, and the ΔE(*O) value was larger for this case. This phenomenon weakened the OH adsorption [59], which facilitated OH deprotonation and accelerated the conversion from OH to O. Overall, the catalytic activity of single Co site improved in the case of oxygen coverage.

Fig. 6(b) showed the density of states for single Ir site under the surface covered of different amounts of O and OH. These results indicated that the occupied states below the Fermi level involved Ir 5d orbitals. For the single Ir site, its adsorption energy for oxygen intermediates was strong, which was located on the left side of Fig. 5(a). The Ir 5d band centers for each case followed the order, −2.44 eV (Ir/

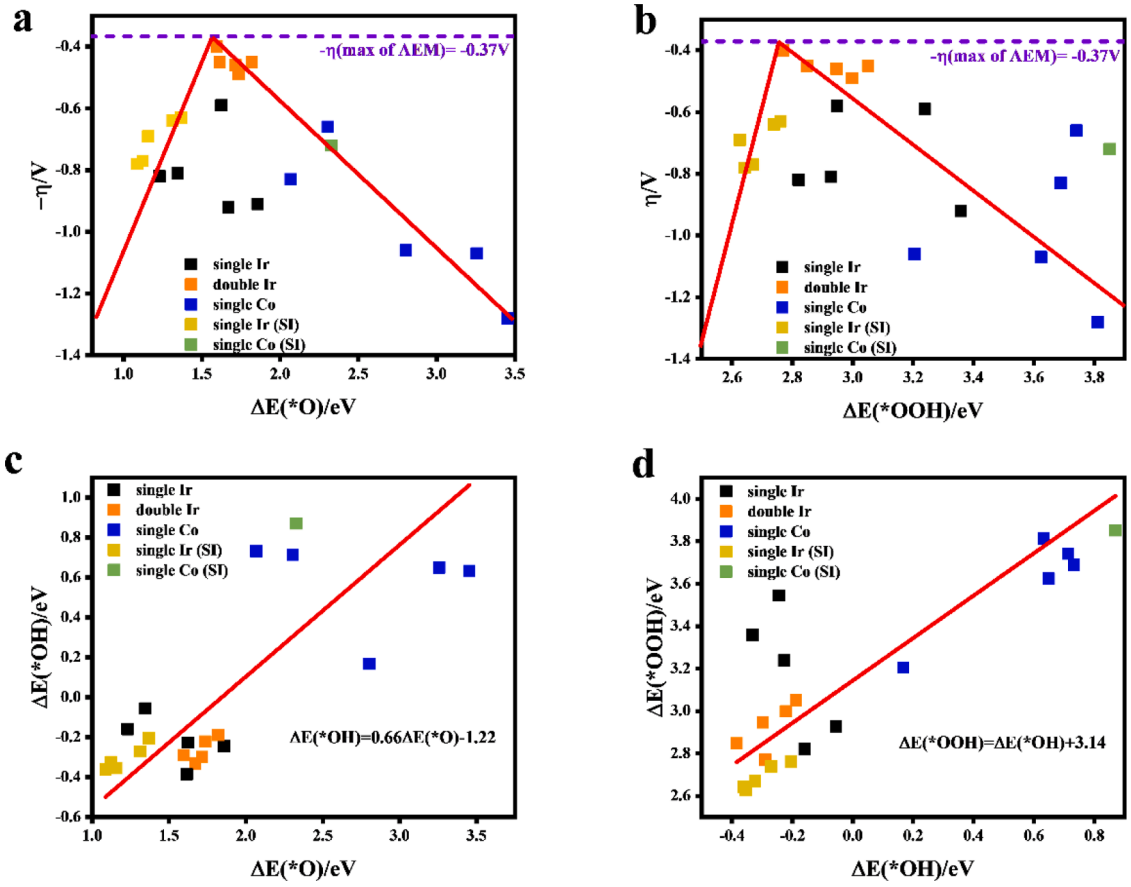


Fig. 5. (a) Volcano diagrams illustrating the relationship between overpotential −η and *O adsorption energy E(*O), (b) volcano diagrams illustrating the relationship between overpotential −η and *OOH adsorption energy E(*OOH). Scaled diagrams relating (c) *OH and *O and (d) *OH and *OOH. Single Ir site, double Ir sites and single Co site under the same coverage conditions were black, orange, and blue respectively. Other cases with single Ir site were shown in yellow and single Co site in green.

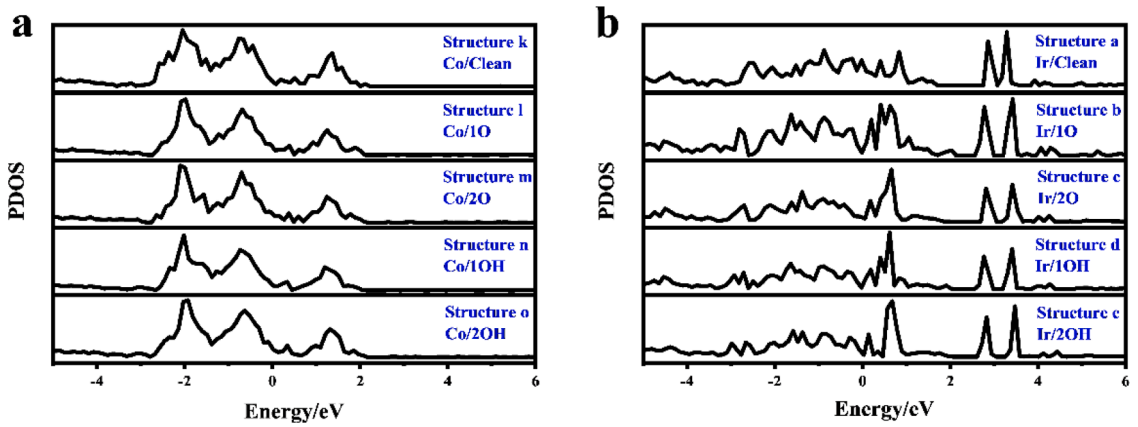


Fig. 6. The partial density of states for (a) single Co site with different surface O and OH covered, and (b) single Ir site with different surface O and OH covered.

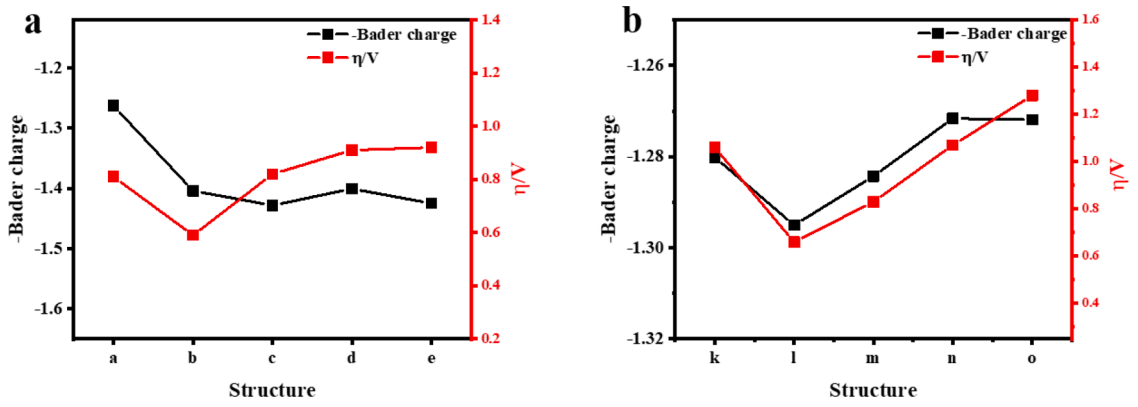


Fig. 7. (a) -Bader charge, and overpotential η curves for single Ir site structures. (b) -Bader charge, and overpotential η curves for single Co site structures.

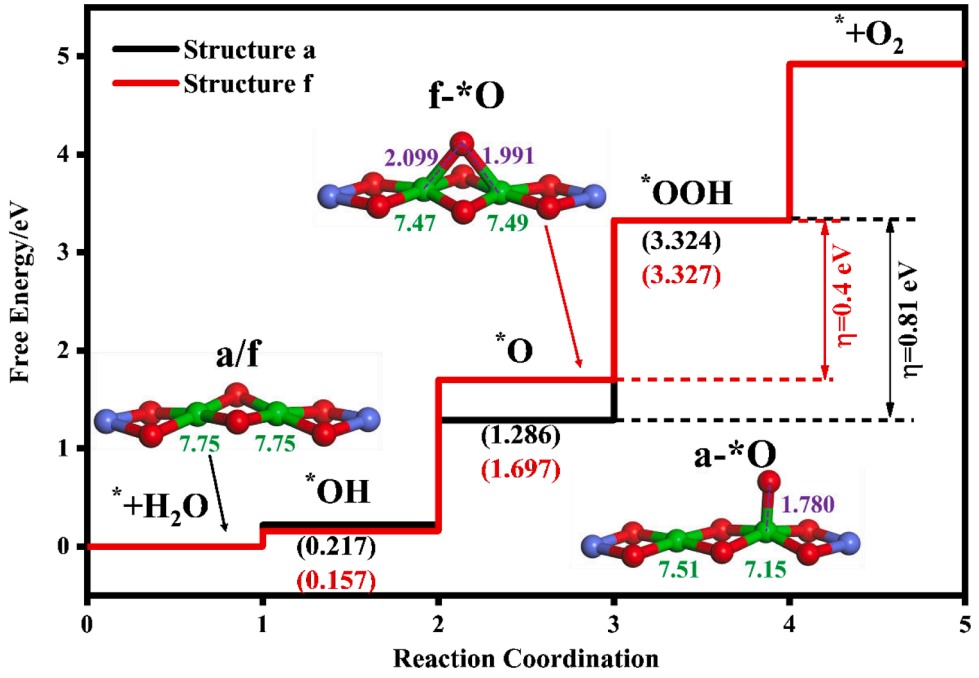


Fig. 8. Comparison of adsorption energies for intermediates between single Ir site and double Ir sites on clean surfaces (structure a and f in Fig. 3). The bond length of Ir-O bond was represented by purple numbers, and the number of valence electrons of Ir atoms was represented by green numbers.

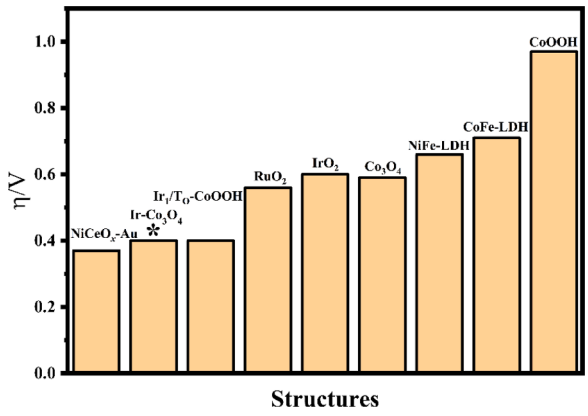


Fig. 9. Comparison of overpotential between this work (*) and other structures.

1OH) < −2.37 eV (Ir/2OH) = −2.37 eV (Ir/2O) < −2.36 eV (Ir/1O) < −2.23 eV (Ir/clean). When the Ir 5d band center was far from the Fermi level, its binding ability to O was significantly weakened that was consistent with the values of ΔE(*O) in Table 3. When the surface was covered with one or two OH, the d-band center moved away from the Fermi level, marking the weakening of the adsorption capacity for oxygen intermediates. When one oxygen covered the surface, the single Ir site had the ideal adsorption energy of oxygen intermediates, which was beneficial to reduce the energy barrier from O to OOH, and the bandgap was narrowed, which was more conducive to the transport of electrons. However, the d-band center moved to the Fermi level when the surface was clean, and it had a large bandgap value, its adsorption energy for oxygen intermediates was enhanced. Therefore, when the surface was covered one oxygen, the single Ir site had an optimistic intermediate adsorption energy, and it had the lowest overpotential.

To further explore the electronic interactions between different adsorbed intermediates and active sites, the Bader charge of single Ir site and single Co site were calculated. The -Bader charge represented the number of transferred valence electrons. The -Bader charge and overpotential curves for "a-e" and "k-m" structures were showed in Fig. 7(a) and Fig. 7(b). For structures with a single Ir site and a single Co site, the overpotential became larger as the number of valence electrons increased and became smaller as the number of valence electrons decreased. So, for single Ir site and single Co site, the surface covered O or OH achieved a lower energy barrier of rate-determining step by reducing the number of valence electrons on the active site.

The OER overpotential was mostly mitigated by the distinctive adsorption manner of double Ir sites and intermediates, rather than by the covered of O and OH intermediates. In order to further study the factors of OER activity of double Ir sites, the energy and adsorption structures of single Ir site and double Ir sites on clean surfaces were compared. Structural and energetic differences arose when intermediates bound Ir sites from different angles, OER reaction intermediates can be stably bound to two Ir atoms at the same time. The difference of free energy in OER four-electron step between single Ir site and double Ir sites was mainly reflected in the adsorption of intermediate O. When the intermediate O was adsorbed on a single Ir site, the length of the Ir-O bond was short, and the number of valence electrons of Ir atom was significantly reduced, corresponding to a stronger O adsorption state (Fig. 8). However, when the intermediate O was adsorbed on two Ir sites at the same time, due to the steric hindrance effect, the bond length of the two Ir-O bonds was longer, resulting in a slight change in the number of Ir valence electrons, weakening the adsorption of O and forming a more ideal adsorption state. Overall, Ir doping with steric structure can reduce the overpotential of OER reaction by adjusting the adsorption energy of oxygen intermediates, which was consistent with previous experimental studies [23]. The results showed that Ir replaced two adjacent Co(III) sites to form a short-range ordered doped structure that had much higher intrinsic activity than single Ir site. Furthermore, the optimum overpotential in this work was compared with that in other reported catalysts, RuO_2, IrO_2 and so on, as shown in Fig. 9 [26,27,52,60-64]. It was found that the double Ir sites on the surface of short-range ordered Ir doping Co_3O_4 had a remarkable OER performance.

It was well known that solvation effect was important for OER, and finally, we considered the effect of solvent on the overpotential of the above "a-o" structures. Table S1 presented the solvation energies "E (sol)" and solvation adsorption energies "ΔE(sol)" for the three active sites under the same coverage. Zhao et al. [65] investigated the OER performance of a two-dimensional $TM_3(HITP)_2$ (TM: transition metal, HITP: 2,3,6,7,10,11-hexaiminotriphenylene functional group) catalyst with and without solvation effects. The calculation showed that the solvent effect will increase the overpotential by more than 0.3 V. Compared with the vacuum, the adsorption energy of OH,O and OOH on the surface became stronger, which was consistent with most of the previously reported catalyst surfaces [31]. Fig. 10 showed -η curves for three active sites in same coverage conditions without and with solvation effect. The order of overpotentials in vacuum at the same site was similar with that in solvent. However, the overpotential of the double Ir sites appeared elevated by about 0.3 V, the reason was that the adsorption energy of the intermediate deviated from the original optimistic state under the influence of solvation.

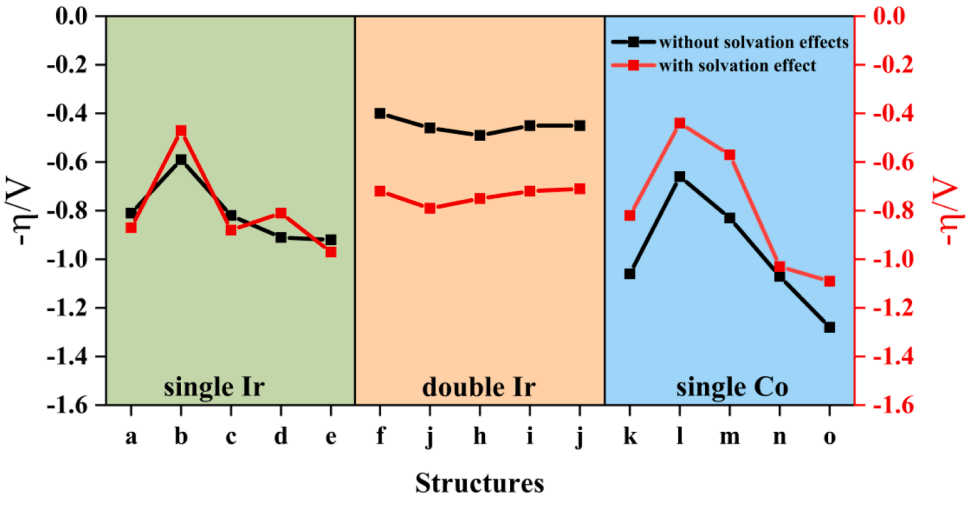


Fig. 10. -η curves for single Ir site, double Ir sites and single Co site in conditions without and with solvation effect.

The solvation energy depended on the adsorption site and the adsorption strength of the intermediate, weaker adsorption states may had a more negative solvation energy [66]. In the double Ir sites structures, the intermediate was bound at the bridge site, and its adsorption was weaker than that of the single Ir site. The oxygen adsorbed at the double Ir sites had a more negative solvation energy as shown in Fig.S1. In addition, O exhibited a larger solvation energy than OH and OOH, which was similar to the previous report [50]. This may be due to the shielding effect of the solvent on the Coulomb interaction between the intermediates and the catalyst surface, which was depended on the type of intermediates and the binding sites [65]. In general, the differences in intermediate types and adsorption sites were the main reasons for the solvation affecting the OER performance. Despite this, the double Ir sites still exhibited fine OER performance in the solvated phase and was virtually unaffected by O or OH covered.

Conclusions

In this work, first-principles computations were performed to investigate the thermodynamics of the OER reaction at three types of metal sites on the surface of a Co_3O_4 electrocatalyst with short-range ordered Ir doping under various intermediate coverage conditions. First, we calculated the adsorption energies and free energies of different structures configurations (single Ir site, double Ir sites, and single Co site) with various intermediates of the four-electron process. Under identical conditions, the catalytic activity of these three types of active sites followed the order: double Ir sites < single Ir site < single Co site. These results confirmed the important role of this unique Ir doping mode in promoting the OER on Co_3O_4 substrate. The partial densities of states and Bader charges of the Ir and Co elements were also calculated and analyzed. For a single active site optimizing the active site's adsorption of intermediates (OH, O, and OOH) was crucial to reducing the overpotential. Meanwhile, surface coverage with oxygen or hydroxyl moieties can tune the adsorption energy and rate-determining step via interactions with electrons at the various active sites. As to double Ir sites, the unique geometry of short-range ordered Ir changed the adsorption structure and adsorption energy of oxygen intermediates compared to isolated Ir atoms, making its surface exhibit excellent OER activity and was hardly affected by surface O and OH covered. The solvation effect induced change in the overpotential depending on the adsorption patterns of the intermediates. The double Ir sites still had a fine OER performance in the solvent phase. The insights discussed herein provide a basis for designing OER catalytic materials with superior performances.

CRediT authorship contribution statement

Zhuang Zhang: Conceptualization, Methodology, Investigation, Software, Data curation, Writing – original draft. **Guoying Tan:** Formal analysis. **Anuj Kumar:** Formal analysis. **Hai Liu:** Software, Validation. **Xue Yang:** Software, Validation. **Wenqin Gao:** Formal analysis. **Lu Bai:** Visualization, Investigation. **Huaiqiu Chang:** Visualization, Investigation. **Yun Kuang:** Writing – review & editing. **Yaping Li:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Xiaoming Sun:** Writing – review & editing.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled. The authors declared that there is no conflict of interest.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the Beijing Natural Science Foundation [grant number Z210016]. We also thank Suzanne Adam, PhD, from Liwen Bianji (Edanz) (https://www.liwenbianji.cn/) for editing the English text of a draft of this manuscript.

References

[1] J. Kibsgaard, I. Chorkendorff, Considerations for the scaling-up of water splitting catalysts, Nature Energy 4 (2019) 430–433, https://doi.org/10.1038/s41560-019-0407-1.
[2] H.A. Gasteiger, N.M. Markovic, Chemistry. Just a dream–or future reality? Science 324 (2009) 48–49, https://doi.org/10.1126/science.1172083.
[3] K. Czelej, J.C. Colmenares, K. Jabłczyńska, K. Ćwieka, Ł. Werner, L. Gradoń, Sustainable hydrogen production by plasmonic thermophotocatalysis, Catal. Today 380 (2021) 156–186, https://doi.org/10.1016/j.cattod.2021.02.004.
[4] S. Chu, A. Majumdar, Opportunities and challenges for a sustainable energy future, Nature 488 (2012) 294–303, https://doi.org/10.1038/nature11475.
[5] M. Carmo, D.L. Fritz, J. Mergel, D. Stolten, A comprehensive review on PEM water electrolysis, Int. J. Hydrogen Energy 38 (2013) 4901–4934, https://doi.org/10.1016/j.ijhydene.2013.01.151.
[6] S. Park, Y.Y. Shao, J. Liu, Y. Wang, Oxygen electrocatalysts for water electrolyzers and reversible fuel cells: status and perspective, Energy Environ. Sci. 5 (2012) 9331–9344, https://doi.org/10.1039/c2ee22554a.
[7] H. Dau, C. Limberg, T. Reier, M. Risch, S. Roggan, P. Strasser, The mechanism of water oxidation: from electrolysis via homogeneous to biological catalysis, ChemCatChem 2 (2010) 724–761, https://doi.org/10.1002/cctc.201000126.
[8] L. Yang, G. Yu, X. Ai, W. Yan, H. Duan, W. Chen, X. Li, T. Wang, C. Zhang, X. Huang, J.S. Chen, X. Zou, Efficient oxygen evolution electrocatalysis in acid by a perovskite with face-sharing IrO6 octahedral dimers, Nat. Commun. 9 (2018) 5236, https://doi.org/10.1038/s41467-018-07678-w.
[9] X. Liu, S. Xi, H. Kim, A. Kumar, J. Lee, J. Wang, N.Q. Tran, T. Yang, X. Shao, M. Liang, M.G. Kim, H. Lee, Restructuring highly electron-deficient metal-metal oxides for boosting stability in acidic oxygen evolution reaction, Nat. Commun. 12 (2021) 5676, https://doi.org/10.1038/s41467-021-26025-0.
[10] A. Marshall, B. Børresen, G. Hagen, M. Tsypkin, R. Tunold, Preparation and characterisation of nanocrystalline IrxSn1−xO2 electrocatalytic powders, Mater. Chem. Phys. 94 (2005) 226–232, https://doi.org/10.1016/j.matchemphys.2005.04.039.
[11] Johnson Matthew Price Charts. www.platinum.matthey.com/prices/price-charts#.
[12] S. Hao, H. Sheng, M. Liu, J. Huang, G. Zheng, F. Zhang, X. Liu, Z. Su, J. Hu, Y. Qian, L. Zhou, Y. He, B. Song, L. Lei, X. Zhang, S. Jin, Torsion strained iridium oxide for efficient acidic water oxidation in proton exchange membrane electrolyzers, Nat. Nanotechnol. 16 (2021) 1371–1377, https://doi.org/10.1038/s41565-021-00986-1.
[13] B. Zhang, X. Zheng, O. Voznyy, R. Comin, M. Bajdich, M. García-Melchor, L. Han, J. Xu, M. Liu, L. Zheng, Homogeneously dispersed multimetal oxygen-evolving catalysts, Science 352 (2016) 333–337, https://doi.org/10.1126/science.aaf1525.
[14] K. Fan, H. Chen, Y. Ji, H. Huang, P.M. Claesson, Q. Daniel, B. Philippe, H. Rensmo, F. Li, Y. Luo, L. Sun, Nickel-vanadium monolayer double hydroxide for efficient electrochemical water oxidation, Nat. Commun. 7 (2016) 11981, https://doi.org/10.1038/ncomms11981.
[15] K.X. Wang, X.Y. Wang, Z.J. Li, B. Yang, M. Ling, X. Gao, J.G. Lu, Q.R. Shi, L.C. Lei, G. Wu, Y. Hou, Designing 3d dual transition metal electrocatalysts for oxygen evolution reaction in alkaline electrolyte: beyond oxides, Nano Energy 77 (2020), 105162, https://doi.org/10.1016/j.nanoen.2020.105162.
[16] M. Xu, X. Zhang, Y. Liu, X. Zhao, Y. Liu, R. Wu, J. Wang, Designed single atom bifunctional Electrocatalysts for overall water splitting: 3d transition metal atoms doped Borophene nanosheets, ChemPhysChem 21 (2020) 2651–2659, https://doi.org/10.1002/cphc.202000692.
[17] W.X. Zhang, L. Cui, J.Q. Liu, Recent advances in cobalt-based electrocatalysts for hydrogen and oxygen evolution reactions, J. Alloys Compd. 821 (2020), 153542, https://doi.org/10.1016/j.jallcom.2019.153542.
[18] K.-.L. Yan, J.-.F. Qin, J.-.H. Lin, B. Dong, J.-.Q. Chi, Z.-.Z. Liu, F.-.N. Dai, Y.-.M. Chai, C.-.G. Liu, Probing the active sites of Co3O4 for the acidic oxygen evolution reaction by modulating the Co2+/Co3+ ratio, J. Mater. Chem. A 6 (2018) 5678–5686, https://doi.org/10.1039/c8ta00070k.
[19] The Institute for Rare Earths and Metals, DOI: https://en.institut-seltene-erden.de/strategic-metals-prices-in-december/.
[20] J. Yu, F.A. Garces-Pineda, J. Gonzalez-Cobos, M. Pena-Diaz, C. Rogero, S. Gimenez, M.C. Spadaro, J. Arbiol, S. Barja, J.R. Galan-Mascaros, Sustainable oxygen evolution electrocatalysis in aqueous 1M H2SO4 with earth abundant nanostructured Co3O4, Nat. Commun. 13 (2022) 4341, https://doi.org/10.1038/s41467-022-32024-6.
[21] A.L. Bhatti, A. Tahira, A. Gradone, R. Mazzaro, V. Morandi, U. aftab, M.I. Abro, A. Nafady, K. Qi, A. Infantes-Molina, A. Vomiero, Z.H. Ibupoto, Nanostructured

Co3O4 electrocatalyst for OER: the role of organic polyelectrolytes as soft templates, Electrochim. Acta 398 (2021), 139338, https://doi.org/10.1016/j.electacta.2021.139338.
[22] C. Cai, M. Wang, S. Han, Q. Wang, Q. Zhang, Y. Zhu, X. Yang, D. Wu, X. Zu, G. E. Sterbinsky, Z. Feng, M. Gu, Ultrahigh oxygen evolution reaction activity achieved using Ir single atoms on amorphous CoOx Nanosheets, ACS Catal. 11 (2020) 123–130, https://doi.org/10.1021/acscatal.0c04656.
[23] J. Shan, C. Ye, S. Chen, T. Sun, Y. Jiao, L. Liu, C. Zhu, L. Song, Y. Han, M. Jaroniec, Y. Zhu, Y. Zheng, S.-.Z. Qiao, Short-range ordered iridium single atoms integrated into cobalt oxide spinel structure for highly efficient Electrocatalytic water oxidation, J. Am. Chem. Soc. 143 (2021) 5201–5211, https://doi.org/10.1021/jacs.1c01525.
[24] J. Rossmeisl, A. Logadottir, J.K. Nørskov, Electrolysis of water on (oxidized) metal surfaces, Chem Phys 319 (2005) 178–184, https://doi.org/10.1016/j.chemphys.2005.05.038.
[25] J.K. Nørskov, J. Rossmeisl, A. Logadottir, L. Lindqvist, J.R. Kitchin, T. Bligaard, H. Jonsson, Origin of the overpotential for oxygen reduction at a fuel-cell cathode, J. Phys. Chem. B 108 (2004) 17886–17892, https://doi.org/10.1021/jp047349j.
[26] P. Zhang, Y. Dong, Y. Kou, Z. Yang, Y. Li, X. Sun, First-principles study of oxygen evolution reaction on the oxygen-containing species covered CoII-Exposing Co3O4 (100) SURFACE, Catal Lett. 145 (2015) 1169–1176, https://doi.org/10.1007/s10562-015-1512-9.
[27] X. Zhao, X. Xiong, X. Duan, Y. Xu, Y. Li, First-principles study of oxygen evolution reaction on Ni3Fe-layered double hydroxides surface with different oxygen coverage, Mol. Catal. 490 (2020), 110957, https://doi.org/10.1016/j.mcat.2020.110957.
[28] G. Tan, X. Zhao, Z. Zhang, X. Yang, J. Chen, L. Xu, Y. Li, Y. Kuang, First-principles study of the oxygen evolution reaction on Ni3Fe-layered double hydroxides surfaces with varying sulfur coverage, Mol. Catal. 519 (2022), 112116, https://doi.org/10.1016/j.mcat.2022.112116.
[29] J.A. Gauthier, C.F. Dickens, L.D. Chen, A.D. Doyle, J.K. Nørskov, Solvation effects for oxygen evolution reaction catalysis on IrO2(110), J. Phys. Chem. C 121 (2017) 11455–11463, https://doi.org/10.1021/acs.jpcc.7b02383.
[30] S.K. Iyemperumal, N.A. Deskins, Evaluating solvent effects at the Aqueous/Pt(111) Interface, ChemPhysChem 18 (2017) 2171–2190, https://doi.org/10.1002/cphc.201700162.
[31] Q. Zhang, A. Asthagiri, Solvation effects on DFT predictions of ORR activity on metal surfaces, Catal. Today 323 (2019) 35–43, https://doi.org/10.1016/j.cattod.2018.07.036.
[32] X. Liu, G. Qiu, X. Li, Shape-controlled synthesis and properties of uniform spinel cobalt oxide nanocubes, Nanotechnology 16 (2005) 3035–3040, https://doi.org/10.1088/0957-4484/16/12/051.
[33] F. Kapteijn, J. Rodriguez-Mirasol, J.A. Moulijn, Heterogeneous catalytic decomposition of nitrous oxide, Appl. Catal. B: Environ. 9 (1996) 25–64, https://doi.org/10.1016/0926-3373(96)90072-7.
[34] A. Montoya, B.S. Haynes, Periodic density functional study of Co3O4 surfaces, Chem. Phys. Lett. 502 (2011) 63–68, https://doi.org/10.1016/j.cplett.2010.12.015.
[35] F. Zasada, W. Piskorz, S. Cristol, J.-.F. Paul, A. Kotarba, Z. Sojka, Periodic density functional theory and atomistic thermodynamic studies of cobalt spinel nanocrystals in wet environment: molecular interpretation of water adsorption equilibria, J. Phys. Chem. C 114 (2010) 22245–22253, https://doi.org/10.1021/jp109264b.
[36] R.-.Z. Zhang, M.J. Reece, High-Entropy Ceramics, Encyclopedia of Materials: Metals and Alloys, 2022, pp. 308–317, https://doi.org/10.1016/b978-0-12-819726-4.00070-3.
[37] I.S. Braude, V.I. Pecherskaya, D.N. Bolshutkin, M.M. Chernik, Short-Range Order in FeCrNi Alloys, Crystal Res. Technol. 21 (1986) 253–258, https://doi.org/10.1002/crat.2170210217.
[38] K. Xia, P. Nan, S. Tan, Y. Wang, B. Ge, W. Zhang, S. Anand, X. Zhao, G.J. Snyder, T. Zhu, Short-range order in defective half-Heusler thermoelectric crystals, Energy Environ. Sci. 12 (2019) 1568–1574, https://doi.org/10.1039/c8ee03654c.
[39] Y. Xu, F. Zhang, T. Sheng, T. Ye, D. Yi, Y. Yang, S. Liu, X. Wang, J. Yao, Clarifying the controversial catalytic active sites of Co3O4 for the oxygen evolution reaction, J. Mater. Chem. A 7 (2019) 23191–23198, https://doi.org/10.1039/c9ta08379k.
[40] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, Comput. Mater. Sci. 6 (1996) 15–50, https://doi.org/10.1016/0927-0256(96)00008-0.
[41] G. Kresse, J. Furthmuller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys Rev B Condens Matter 54 (1996) 11169–11186, https://doi.org/10.1103/physrevb.54.11169.
[42] P.E. Blöchl, Projector augmented-wave method, Phys. Rev. B 50 (1994) 17953, https://doi.org/10.1103/physrevb.50.17953.
[43] G. Kresse, D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, Phys. Rev. B 59 (1999) 1758, https://doi.org/10.1103/PhysRevB.59.1758.
[44] J.E. Saal, Y. Wang, S. Shang, Z.-.K. Liu, Thermodynamic properties of Co3O4 and Sr6Co5O15 from first-principles, Inorg Chem 49 (2010) 10291–10298, https://doi.org/10.1021/ic100835a.
[45] L. Wang, T. Maxisch, G. Ceder, A first-principles approach to studying the thermal stability of oxide cathode materials, Chem. Mater. 19 (2007) 543–552, https://doi.org/10.1021/cm0620943.
[46] P. Li, X. Zhao, X. Duan, Y. Li, Y. Kuang, X. Sun, A multiphase nickel iron sulfide hybrid electrode for highly active oxygen evolution, Sci. China Mater. 63 (2019) 356–363, https://doi.org/10.1007/s40843-019-1215-9.
[47] Z. Cai, D. Zhou, M. Wang, S.M. Bak, Y. Wu, Z. Wu, Y. Tian, X. Xiong, Y. Li, W. Liu, S. Siahrostami, Y. Kuang, X.Q. Yang, H. Duan, Z. Feng, H. Wang, X. Sun, Introducing Fe(2+) into nickel-iron layered double hydroxide: local structure modulated water oxidation activity, Angew. Chem. Int. Ed Engl. 57 (2018) 9392–9396, https://doi.org/10.1002/anie.201804881.
[48] M. Bajdich, M. Garcia-Mota, A. Vojvodic, J.K. Norskov, A.T. Bell, Theoretical investigation of the activity of cobalt oxides for the electrochemical oxidation of water, J. Am. Chem. Soc. 135 (2013) 13521–13530, https://doi.org/10.1021/ja405997s.
[49] K.S. Exner, A short perspective of modeling electrode materials in lithium-ion batteries by the ab initio atomistic thermodynamics approach, J. Solid State Electrochem. 22 (2018) 3111–3117, https://doi.org/10.1007/s10008-018-4017-9.
[50] Y. Sha, T.H. Yu, Y. Liu, B.V. Merinov, W.A. Goddard, Theoretical study of solvent effects on the platinum-catalyzed oxygen reduction reaction, J. Phys. Chem. Lett. 1 (2010) 856–861, https://doi.org/10.1021/jz9003153.
[51] S. Hao, M. Liu, J. Pan, X. Liu, X. Tan, N. Xu, Y. He, L. Lei, X. Zhang, Dopants fixation of Ruthenium for boosting acidic oxygen evolution stability and activity, Nat. Commun. 11 (2020) 5368, https://doi.org/10.1038/s41467-020-19212-y.
[52] Z. Zhang, C. Feng, D. Wang, S. Zhou, R. Wang, S. Hu, H. Li, M. Zuo, Y. Kong, J. Bao, J. Zeng, Selectively anchoring single atoms on specific sites of supports for improved oxygen evolution, Nat. Commun. 13 (2022) 2473, https://doi.org/10.1038/s41467-022-30148-3.
[53] J. Xu, D. Wang, Y. Cai, J. Liu, Two-dimensional polycrystalline Co3O4 supported high-number-density metal single atoms and clusters, Microsc. Microanal. 25 (2019) 2210–2211, https://doi.org/10.1017/s1431927619011784.
[54] K. Liu, F. Wang, P. He, T.A. Shifa, Z. Wang, Z. Cheng, X. Zhan, J. He, The role of active oxide species for electrochemical water oxidation on the surface of 3d-Metal phosphides, Adv. Energy Mater. 8 (2018), 1703290, https://doi.org/10.1002/aenm.201703290.
[55] J. Song, C. Wei, Z.F. Huang, C. Liu, L. Zeng, X. Wang, Z.J. Xu, A review on fundamentals for designing oxygen evolution electrocatalysts, Chem. Soc. Rev. 49 (2020) 2196–2214, https://doi.org/10.1039/c9cs00607a.
[56] J. Rossmeisl, Z.W. Qu, H. Zhu, G.J. Kroes, J.K. Nørskov, Electrolysis of water on oxide surfaces, J. Electroanal. Chem. 607 (2007) 83–89, https://doi.org/10.1016/j.jelechem.2006.11.008.
[57] X. Ren, F. Hou, F. Wang, X. Zhang, Q. Wang, Porous CoO-CeO2 heterostructures as highly active and stable electrocatalysts for water oxidation, Int J Hydrogen Energy 43 (2018) 22529–22537, https://doi.org/10.1016/j.ijhydene.2018.10.096.
[58] B. Hammer, J.K. Nørskov, Theoretical Surface Science and Catalysis—Calculations and Concepts, Advances in Catalysis, Academic Press, 2000, pp. 71–129, https://doi.org/10.1016/S0360-0564(02)45013-4.
[59] I.C. Man, H.Y. Su, F. Calle-Vallejo, H.A. Hansen, J.I. Martínez, N.G. Inoglu, J. Kitchin, T.F. Jaramillo, J.K. Nørskov, J. Rossmeisl, Universality in oxygen evolution electrocatalysis on oxide surfaces, ChemCatChem 3 (2011) 1159–1165, https://doi.org/10.1002/cctc.201000397.
[60] Y. Wang, S. Hao, X. Liu, Q. Wang, Z. Su, L. Lei, X. Zhang, Ce-Doped IrO2 Electrocatalysts with Enhanced Performance for Water Oxidation in Acidic Media, ACS Appl. Mater. Interfaces 12 (2020) 37006–37012, https://doi.org/10.1021/acsami.0c00389.
[61] P. Li, M. Wang, X. Duan, L. Zheng, X. Cheng, Y. Zhang, Y. Kuang, Y. Li, Q. Ma, Z. Feng, W. Liu, X. Sun, Boosting oxygen evolution of single-atomic ruthenium through electronic coupling with cobalt-iron layered double hydroxides, Nat. Commun. 10 (2019) 1711, https://doi.org/10.1038/s41467-019-09666-0.
[62] J. He, X. Zhou, P. Xu, J. Sun, Regulating electron redistribution of intermetallic iridium oxide by incorporating Ru for efficient acidic water oxidation, Adv. Energy Mater 11 (2021), https://doi.org/10.1002/aenm.202102883.
[63] J.W.D. Ng, M. García-Melchor, M. Bajdich, P. Chakthranont, C. Kirk, A. Vojvodic, T.F. Jaramillo, Gold-supported cerium-doped NiOx catalysts for water oxidation, Nature Energy 1 (2016) 16053, https://doi.org/10.1038/nenergy.2016.53.
[64] A. Zagalskaya, V. Alexandrov, Role of defects in the interplay between adsorbate evolving and lattice oxygen mechanisms of the oxygen evolution reaction in RuO2 and IrO2, ACS Catal 10 (2020) 3650–3657, https://doi.org/10.1021/acscatal.9b05544.
[65] J. Wang, Y. Fan, S. Qi, W. Li, M. Zhao, Bifunctional HER/OER or OER/ORR catalytic activity of Two-dimensional TM3(HITP)2 with TM = Fe–Zn, J. Phys. Chem. C 124 (2020) 9350–9359, https://doi.org/10.1021/acs.jpcc.0c01143.
[66] C.M. Gray, K. Saravanan, G. Wang, J.A. Keith, Quantifying solvation energies at solid/liquid interfaces using continuum solvation methods, Mol. Simul. 43 (2017) 420–427, https://doi.org/10.1080/08927022.2016.1273525.